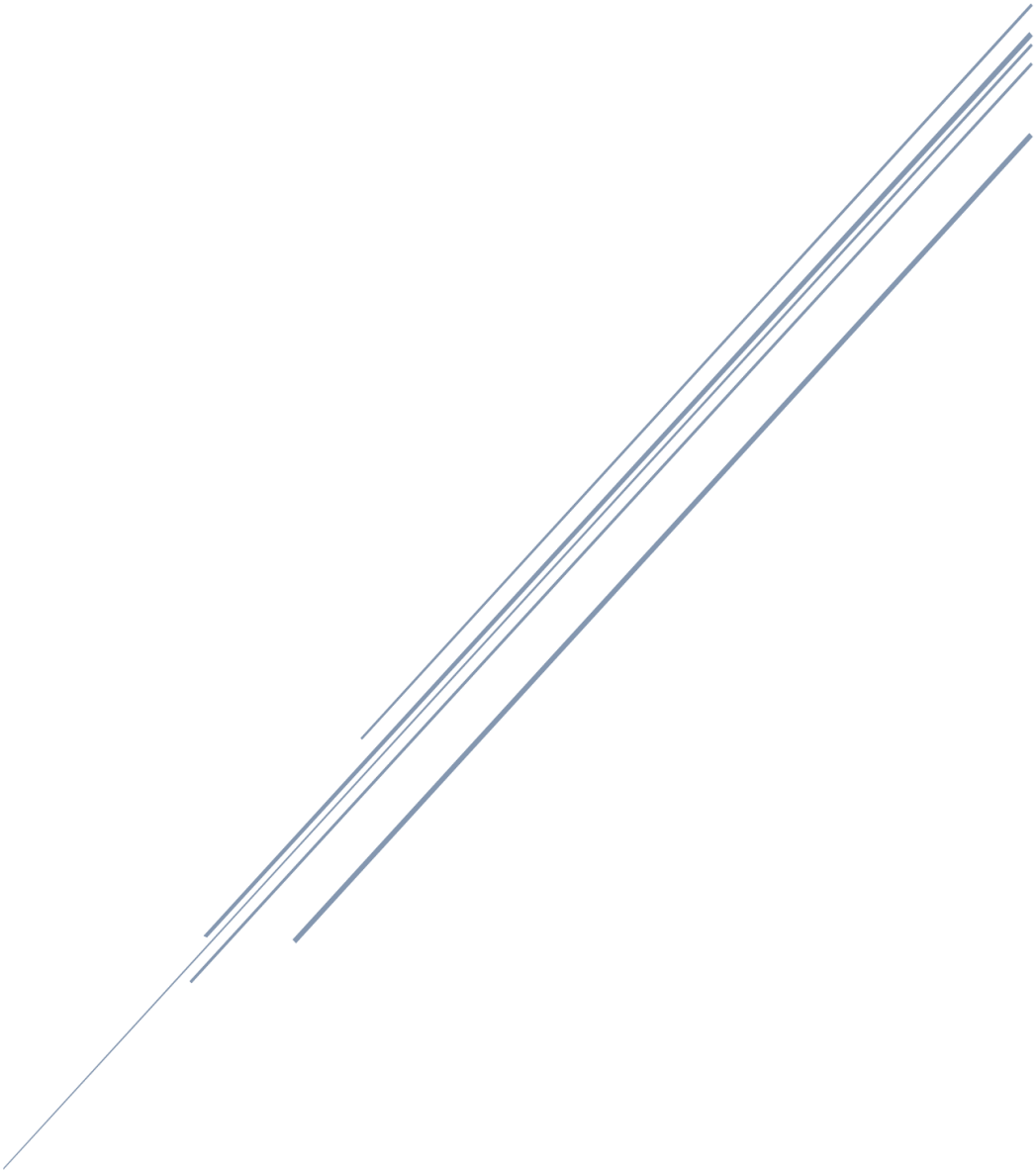
INTERNSHIP DAILY REPORT

Day 3 – Linux Admin and Networking Basics

CyberShelter Cybersecurity Internship | Week 1



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1. Objective

The objective of Day 3 was to continue learning Linux administration, covering networking, services, logs and cron, and to practise network configuration, connectivity testing, port checking and log reading in Kali Linux. The hands-on task was to identify the IP address, gateway and DNS server of the system, and to explain the difference between TCP and UDP.

2. Lab Environment

- **Host Machine:** Windows PC
- **Virtualisation Software:** Oracle VirtualBox (NAT network)
- **Guest OS:** Kali Linux (64-bit), VM "Kali Lab"
- **User Account:** shameem (normal user with sudo access)

3. Topics Covered

3.1 Networking

Networking lets computers communicate and share data. In Linux, admins manage networking by checking IP addresses, controlling access with a firewall, managing servers remotely with SSH, and watching network traffic for problems or attacks.

- **Network Basics:** An **IP address** is a unique number for each device, the **gateway** is the router that sends traffic out to other networks, **DNS** turns names like google.com into IP addresses (stored in `/etc/resolv.conf`), and **ports** are numbered doors used by services, for example 22 SSH, 53 DNS, 80 HTTP and 443 HTTPS.
- **Firewall:** A firewall controls which network traffic is allowed in or blocked. CentOS uses **firewalld** with zones such as public and drop, while Kali uses **ufw** or **iptables**.
- **Remote Management (SSH):** SSH gives an encrypted command line to a remote server on port 22. Good practice is to set `PermitRootLogin no` and log in as a normal user first.
- **Traffic Monitoring:** Tools like `nload`, `iftop`, `tcpdump` and Wireshark show how much data flows through the network and help find slowdowns or suspicious activity.

3.2 Services

A service, also called a daemon, is a program that runs in the background, such as SSH, a web server or logging. In Linux, services are managed by **systemd**, the first process (PID 1) started at boot, using the `systemctl` command.

- **Controlling Services:** `systemctl status ssh` shows whether a service is running, and `sudo systemctl start / stop / restart ssh` controls it.
- **Boot Management:** `sudo systemctl enable ssh` starts a service automatically at boot, and `systemctl list-units --type=service` lists all services.
- **Security Point:** Every running service can be a way in for attackers, so services that are not needed should be disabled to reduce the attack surface.

3.3 Logs

Logs are records of everything that happens on a system: logins, errors, services and kernel events. In modern Linux, **journald** collects the logs and `journalctl` reads them. Logs are one of the most important sources for troubleshooting and security investigations.

- **Log Locations:** `/var/log/` is the main log folder. `/var/log/syslog` (Debian) and `/var/log/messages` (CentOS) hold general logs, and `/var/log/auth.log` holds logins and sudo use.
- **journalctl:** `journalctl -b` shows logs from the current boot, `-u ssh` shows logs for one service, `-p 4` shows warnings and errors, and `--since "1 hour ago"` filters by time.
- **Good Practice:** Keep the correct time zone with `timedatectl`, and watch for repeated failed logins, which can be a sign of a brute-force attack.

3.4 Cron (Scheduled Tasks)

Cron is a Linux service that runs commands automatically at set times, for example daily backups, cleaning old logs or running scripts. Each user's scheduled jobs are stored in a file called a **crontab**.

- **Commands:** `crontab -e` edits jobs, `crontab -l` lists them and `crontab -r` removes them.
- **Format:** `minute hour day month weekday command`. For example, `0 2 * * * /home/shameem/backup.sh` runs every day at 2:00 AM, and `*/5 * * * *` runs every 5 minutes.
- **Security Point:** Attackers sometimes add hidden cron jobs to keep access to a hacked system (persistence), so admins regularly check `crontab -l` and `/etc/crontab`.

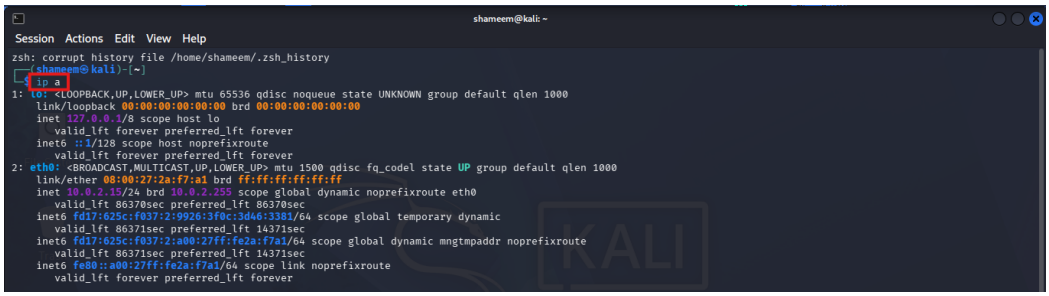
4. Tasks Performed

All commands were run in the Kali Linux terminal as the normal user shameem. My own website, mohamedshameem.com, was used as the target for the connectivity tests.

4.1 Network Configuration

4.1.1 IP Address

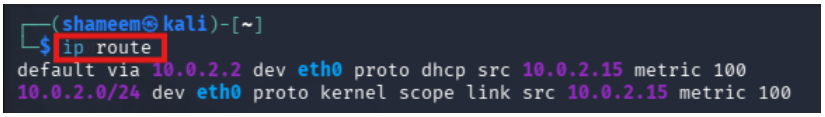
- **Command Executed:** `ip a`
- **Purpose:** Shows all network interfaces with their IP and MAC addresses.
- **Observation:** Two interfaces are listed: **lo** (loopback, 127.0.0.1) and **eth0**, the main network card, which has the IP address **10.0.2.15/24** and is in the UP state.



Network interfaces and IP addresses.

4.1.2 Routing Table

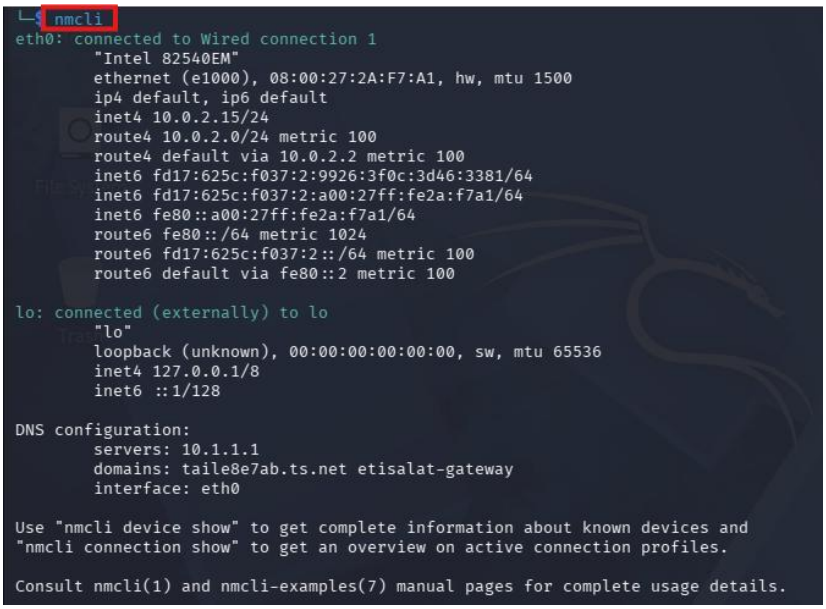
- **Command Executed:** `ip route`
- **Purpose:** Shows how traffic is routed and which gateway is used.
- **Observation:** The line **default via 10.0.2.2 dev eth0** shows that the default gateway is **10.0.2.2**, which is the VirtualBox NAT router. The local network is 10.0.2.0/24.



Routing table and default gateway.

4.1.3 Network Manager

- **Command Executed:** `nmcli`
- **Purpose:** Shows the network connection status, adapter details and DNS settings.
- **Observation:** eth0 is connected to "Wired connection 1" using an **Intel 82540EM** adapter. The **DNS configuration** section shows the DNS server **10.1.1.1**.



Connection details and DNS configuration.

4.2 Connectivity Testing

4.2.1 Ping Test

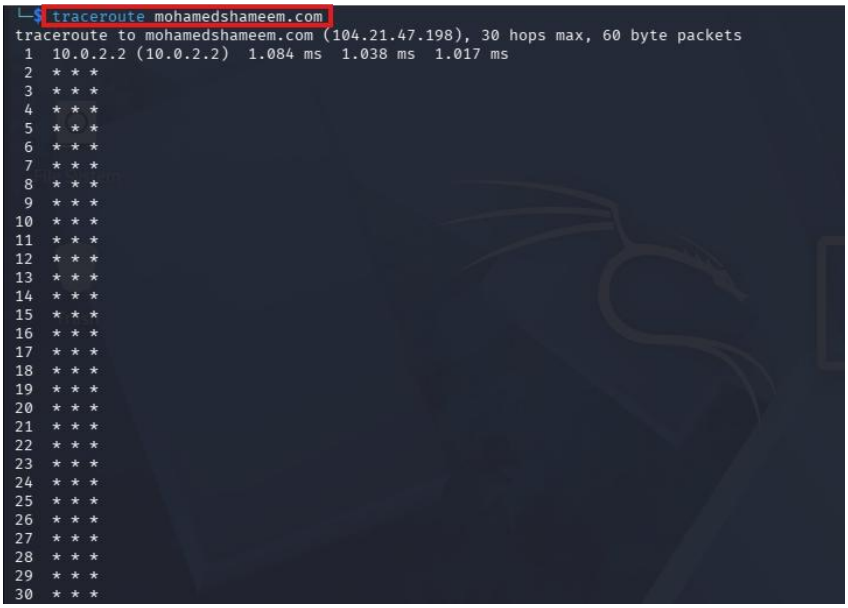
- **Command Executed:** `ping mohamedshameem.com`
- **Purpose:** Checks whether a host is reachable and measures the response time.
- **Observation:** The domain resolved to **172.67.172.111** and replied with an average time of about **109 ms**. 9 packets were sent and 7 received (about 22% packet loss). The command was stopped with **Ctrl + C**.



Ping test to mohamedshameem.com.

4.2.2 Traceroute

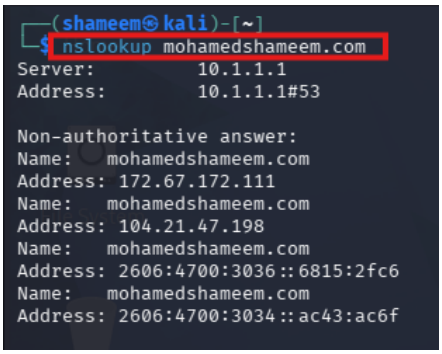
- **Command Executed:** `traceroute mohamedshameem.com`
- **Purpose:** Shows the path (hops) that packets take to reach the destination.
- **Observation:** The first hop is the gateway **10.0.2.2** (about 1 ms). The remaining hops show `* * *` because the VirtualBox NAT network and routers on the way did not reply to traceroute packets.



Traceroute to mohamedshameem.com.

4.2.3 DNS Lookup

- **Command Executed:** `nslookup mohamedshameem.com`
- **Purpose:** Finds the IP addresses of a domain using DNS.
- **Observation:** The DNS server **10.1.1.1** answered with two IPv4 addresses (**172.67.172.111** and **104.21.47.198**) and two IPv6 addresses. The answer is marked "Non-authoritative" because it came from a DNS resolver, not from the domain's own name server.



DNS lookup for mohamedshameem.com.

4.3 Ports & Services

4.3.1 Listening Ports

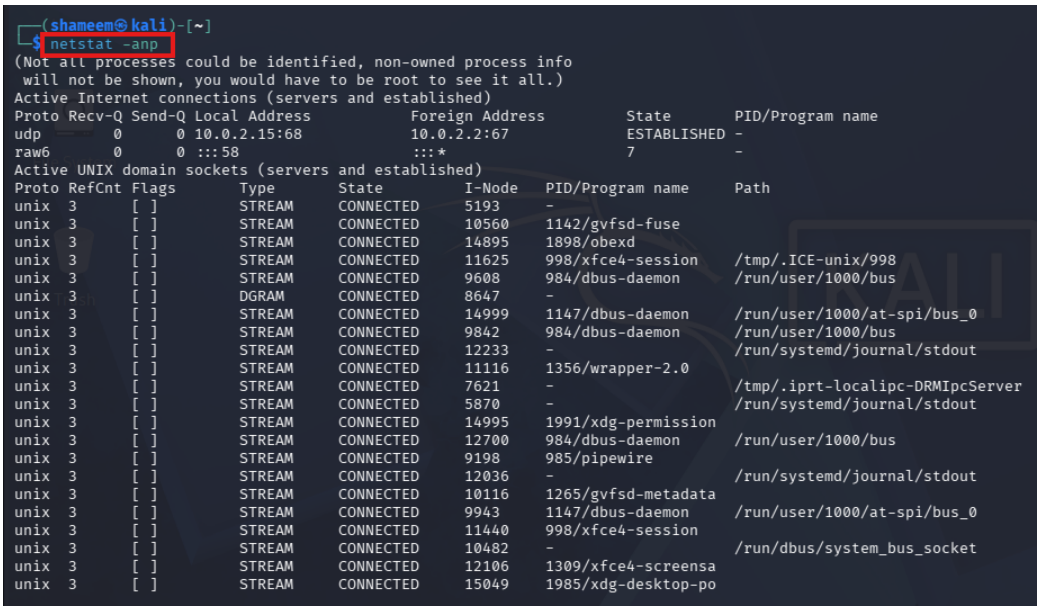
- **Command Executed:** `ss -tulnp`
- **Purpose:** Lists all TCP and UDP ports that are open and listening on the system.
- **Observation:** Only the column headers are shown, which means **no services are listening** for incoming connections. This is a good security state, because there are no open ports for an attacker to connect to.



Listening TCP and UDP ports.

4.3.2 Network Connections

- **Command Executed:** `netstat -anp`
- **Purpose:** Shows all active network connections and the programs using them.
- **Observation:** There is one active UDP connection from **10.0.2.15:68** to **10.0.2.2:67**, which is the DHCP client getting its IP address from the gateway. The rest are local UNIX sockets used by system programs like dbus and xfce4-session.

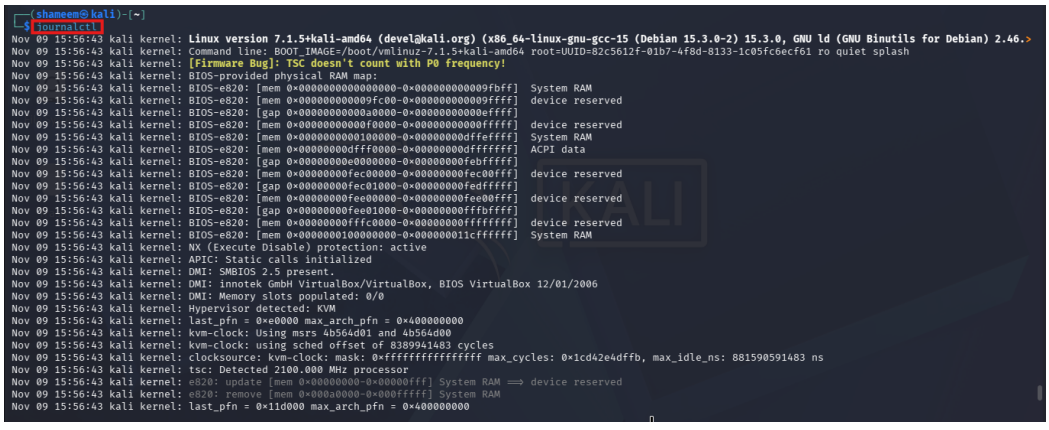


Active network connections.

4.4 Logs

4.4.1 System Logs

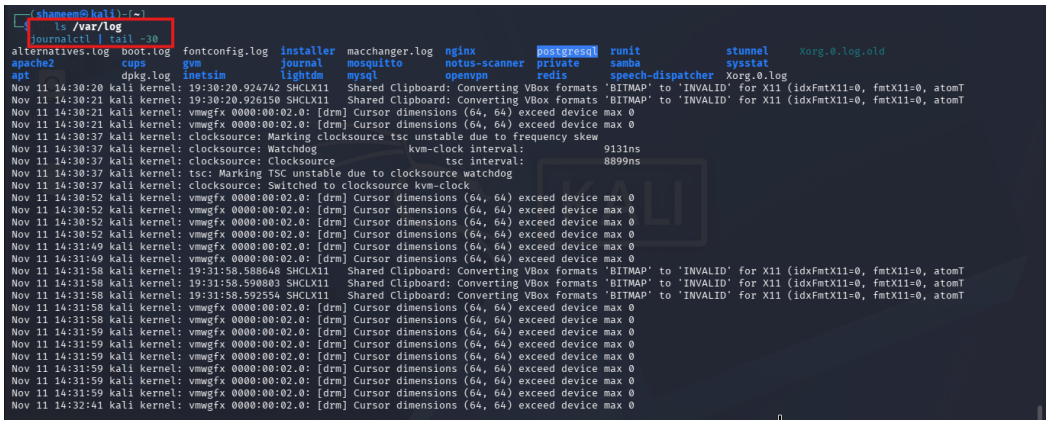
- **Command Executed:** `journalctl`
- **Purpose:** Shows the system logs collected by systemd-journald.
- **Observation:** The logs start from boot and show kernel messages such as the Linux version, BIOS memory map, **DMI: innotek GmbH VirtualBox** and **Hypervisor detected: KVM**.



System logs from journalctl.

4.4.2 Log Directory and Recent Logs

- **Command Executed:** `ls /var/log  and  journalctl | tail -30`
- **Purpose:** Lists the log files and shows the most recent log entries.
- **Observation:** The /var/log folder contains logs for services like **apt**, **apache2**, **nginx**, **mysql** and **postgresql**, but **no syslog file**. The last 30 journal entries show recent kernel and VirtualBox shared clipboard messages.



Log directory and recent journal entries.

5. Results

5.1 Network Details

Item	Result
IP Address	10.0.2.15/24 (eth0)
Default Gateway	10.0.2.2 (VirtualBox NAT router)
DNS Server	10.1.1.1
Network Adapter	Intel 82540EM (e1000)
Listening Ports	None

5.2 Difference between TCP and UDP

Feature	TCP	UDP
Connection	Connection-based (3-way handshake)	Connectionless
Reliability	Reliable, resends lost data	Not guaranteed, no resending
Order	Data arrives in order	Data may arrive out of order
Speed	Slower	Faster
Used for	Web (HTTP/HTTPS), email, SSH, file transfer	DNS, video streaming, online gaming, VoIP

TCP 3-way handshake: SYN → SYN-ACK → ACK. The connection is set up before any data is sent. In this practical, the DHCP connection seen in `netstat` (port 67/68) used **UDP**.

6. Key Learnings

- Learned how to find the IP address, gateway and DNS server of a Linux system using `ip a`, `ip route` and `nmcli`.
- Understood how to test connectivity and DNS using `ping`, `traceroute` and `nslookup`.
- Learned to check open ports and connections with `ss` and `netstat`.
- Understood the difference between TCP and UDP and where each is used.

7. Challenges Faced

- **/var/log/syslog not found:** `cat /var/log/syslog` showed "No such file or directory" because Kali uses journald, so the logs were viewed with `journalctl` instead.
- **Traceroute showed only stars:** After the first hop, traceroute showed `* * *` because the VirtualBox NAT network did not reply, but the first hop confirmed the gateway 10.0.2.2.

8. Conclusion & Next Steps

Day 3 showed how a Linux system connects to a network and how services, logs and cron are managed. The hands-on task confirmed the IP address, gateway and DNS server, with no open ports. **Next Steps:** Day 4 will focus on Windows and networking from the attacker's view.